Hand hygiene in hospital wards

Improve compliance without a training module. Project name: Threshold.

Sector: Healthcare, ward operations

Status: concept study, written and designed by Karthik Bindumon, 2026. Numbers are targets or modelled assumptions and are labelled as such.

Method: The Realisation Loop, v1 (see 00-realisation-loop.pdf).

1 Brief

Hospitals train hand hygiene annually, and audits keep finding compliance below target. The WHO's 2009 Guidelines on Hand Hygiene in Health Care reviewed dozens of observational studies and reported an average compliance near 39 percent, with wide variation. Retraining does not move the number for long. This study designs an intervention for one ward that needs no training module at all.

2 Audience and needs analysis

Primary audience: nursing staff and ward attendants entering a general medical ward, 40 to 60 entries per person per shift. What they already know: the WHO Five Moments, taught yearly. What they lack: a cue at the moment the hand is free and the body is already moving. Constraints: no time in the shift for anything, alarm fatigue, resentment of surveillance, patient privacy in rooms.

Needs analysis method for the real project: two shifts of unobtrusive observation, six short interviews at shift end, dispenser consumption logs for the last quarter, and the ward's existing audit sheets.

3 The realisation

> **Nurses know the rule. They miss the moment. The failure is cue timing at the ward door, not knowledge. So the door has to ask, at the second the hand is free.**

Contrarian core: retire the annual module for this ward for the pilot period and spend the budget on the threshold. If the number moves without training, the ward has learned something the training industry would rather it did not.

4 Learning objective (target, Mager form)

By week 8 of the pilot (condition), ward staff (audience) perform hand rub at ward entry (behaviour) on at least 75 percent of covertly observed entries, up from the audited baseline, with no verbal prompting (degree).

5 Design

The Threshold. A pressure tile set into the floor just inside the ward door. Stepping on it lights a short strip path (floor LEDs under a wear layer) to the wall dispenser, 1.5 metres away. The dispenser pump closes a low-voltage contact: a soft wooden click plays from a speaker in the dispenser housing and the tile fades from gold to neutral. The turn is over. A missed entry does nothing; the tile fades on its own after eight seconds. Silence is the only penalty.

Team loop. Twelve consecutive clean entries raise a paper flag in a slot at the nurse station, moved by hand by whoever notices. Analog on purpose: the ward owns the flag, no dashboard owns the ward.

Implicit learning. Nobody is told anything. The floor teaches the timing by being the cue. Within a week the tile-to-pump path becomes a chunked routine.

Artefacts. Tile and strip hardware spec, dispenser contact retrofit, a one-page ward note ('a new tile at the door, nothing to do'), observation sheet, two-question reaction card.

6 Theory map

Source	How it is used
Fogg, Behaviour Model (B = MAP)	The prompt sits at the exact point of highest ability: hand free, body moving, dispenser in reach.
Gollwitzer, implementation intentions	'When I cross the tile, I rub' forms without instruction because cue and act are adjacent in space and time.
Van Gennep; Turner	Thresholds already carry ritual weight (rites of passage, the Japanese genkan). The design borrows existing meaning instead of building new meaning.
Graybiel, habit chunking	A fixed cue, a fixed short routine, an immediate sensory close. Nothing variable, nothing delayed.
Schultz, reward prediction error	Variable reward drives seeking; hygiene needs reliability. Feedback is constant and immediate by design.
Deci and Ryan, SDT	Team flag, no names, no counts per person. Autonomy and relatedness survive; surveillance would end both.

7 Evaluation plan (Kirkpatrick)

Level	Measure	When
1 Reaction	Two-question paper card at shift end (did the tile get in the way; would you keep it).	Weeks 1, 4, 8
2 Learning	Cue-recognition task: a photo of a doorway; mark the moment. Not a knowledge quiz.	Week 0 and 8
3 Behaviour	Covert observation (trained observer, WHO method) and dispenser consumption per patient-day, pilot ward against a matched control ward.	Baseline 2 weeks, then weekly
4 Results	Healthcare-associated infection rate, owned by infection control.	6 months

8 Pilot plan and risks

One ward, eight weeks, matched control ward, two-week baseline. Budget line items: tile and strip hardware, dispenser retrofit, observer hours, two paper artefacts. Risks: novelty effect fading after week 3 (mitigation: the flag loop; measure week 6 to 8 separately); tile perceived as a trip hazard (flush install, non-slip wear layer); staff reading the tile as surveillance (the ward note says plainly that nothing is recorded, and nothing is).

9 Ethics

No individual tracking of any kind. No sound audible beyond two metres. Patient rooms out of scope. The pilot can be stopped by the ward manager on any day without explanation.

10 What I would need from the client

Access to the last four quarters of audit and consumption data, one infection-control contact, two weeks of observer access, and the authority to pause the annual module for the pilot ward.